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(54) Title: **Design and development of a water cleaning bot for all water bodies**

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(57) Abstract:

“Design and Development of a Water Cleaning Bot for all water bodies” **ABSTRACT** The water cleaning bot is a novel solution to aquatic environment maintenance that combines modern waste management and water quality monitoring devices. The bot collects floating waste from water surfaces using a conveyor belt system, which is controlled via an Android interface. A sophisticated segregation system, which employs infrared (IR) and inductive proximity sensors, separates the collected trash into metallic and non-metallic components, which are then placed in specified compartments of a collecting tray. Furthermore, the bot is outfitted with sensors that monitor critical water quality metrics such as total dissolved solids (TDS), pH levels, and turbidity, giving useful data for environmental assessments. The Long Range (LoRa) protocol enables dependable navigation and communication over long distances. This technique provides a long-term and effective solution for maintaining water quality and reducing pollution in aquatic habitats.

FORM 2 THE PATENTS ACT 39 OF 1970 & THE PATENT RULES 2003 COMPLETE SPECIFICATION (SEE SECTIONS 10 & RULE 13)		
1. TITLE OF THE INVENTION “Design and Development of a Water Cleaning Bot for all water bodies”		
2. APPLICANTS (S)		
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3. PREAMBLE TO THE DESCRIPTION		
4. COMPLETE SPECIFICATION The following specification particularly describes the invention and the manner in which it is to be performed		

“Design and Development of a Water Cleaning Bot for all water bodies”

TECHNICAL FIELD

- [1] The present invention relates to the field of environmental engineering and robotics, specially focusing on the automated and manually controlled Water Cleaning Bot.
- [2] It is designed to work on various waterbodies and collect wastes, measure water purification parameters using different sensors and filter the polluted water.
- [3] It addresses the problem of water pollution, where it targets the environmental conservationists, government agencies and communities who are seeking sustainable solutions for cleaning and maintaining water bodies.

BACKGROUND OF THE INVENTION

- [4] Water pollution is a significant environmental concern that has affected water bodies globally. The significant increase in accumulation wastes, chemicals and plastics over the years have intensified the need of efficient and scalable solutions to maintain these environments.
- [5] As urbanization and industrialization accelerates, the volume of pollutants entering water bodies has increased with plastics and organic

wastes being the primary contributors. These pollutants reduce the quality of the water body and makes it unfit for usage in household purposes, agriculture etc.

- [6] Consequently, this creates an urgent need for solutions which can effectively tackle the problem. Advances in robotics and automation technologies promise to provide more tractable solutions to this challenge.
- [7] A water cleaning bot, which embodies a single human-controlled solution, that navigates through water bodies, collecting waste and filtering out contaminants, promises a scalable solution. It could run using the single human intervention, ensuring that the operation of cleaning the water body, becomes repetitive and, hence, regular.
- [8] Such features includes waste detection sensors, water quality detecting sensors, solar-powered systems to boost the sustainability of the product, and thereby making the operation faster and less labour-intensive.

SUMMARY OF THE INVENTION

- [9] The present invention is a manually controlled water cleaning bot which is intended to operate in any of the water bodies, including rivers, lakes, ponds. It is a practical tool for collecting and filtering waste from the water for a cleaner aquatic environment.
- [10] The bot makes possible an efficient real-time operation by a human operator as it integrates specific sensors and waste management features and filtering techniques. This robot also employs an inductive proximity sensor. This allows for the categorization of trash based on

material composition.

- [11] It assesses the content of metallic debris and may differentiate between non-metallic refuse and metallic waste. The waste segregation system enhances the collection process, facilitating the recycling or proper disposal of goods. This mitigates the environmental impact of waste management activities.
- [12] A TDS sensor on the device quantifies the concentration of dissolved substances in the water. This provides critical information regarding water quality and pollutant concentrations. The bot is equipped with a pH sensor that continuously assesses the acidity or alkalinity of the water and identifies the presence of hazardous compounds or impurities.
- [13] The onboard system effectively gathers waste by seizing surface materials like plastics, organic substances, and various contaminants. Using the data received from the inductive proximity sensor, it would discriminate metallic and non-metallic materials.
- [14] This will boost the efficiency of the bot, and hence the activities for better structured rubbish disposal or recycling will take place. The filtering section includes a basic filter mechanism that filters out all microplastics from the water and gives a cleaner supply. This bot is constructed from sturdy and rust-resistant materials, with its goal for long-term employment in the most diverse aquatic environments: from freshwater lakes to salty oceans.
- [15] It's a humanly operated water-cleaning bot that addresses the demand for effective management of water pollution by bringing together human control with the most advanced technologies for waste

segregation and monitoring of water quality. Because of its ability to sort rubbish, monitor real-time water quality, and perform under diverse situations, it has proven to be a vital instrument in the relentless drive to maintain water bodies cleaner and healthier.

DETAILED DESCRIPTION OF THE INVENTION

- [16] The water cleaning bot is a revolutionary method of cleaning the water's surface together with constant monitoring of water quality. It may also segregate the accumulated rubbish from the water surface. It is driven by an esp8266 microcontroller, which connects to a website and delivers data via IoT for checking water quality.
- [17] The sensors used for the water quality monitoring system feature are pH sensor, turbidity sensor and TDS sensor. These sensors are connected to the microcontroller.
- [18] The microcontroller is also connected to an esp32 cam module which is equipped with the object detection model which allows in detecting the object and navigates towards it.
- [19] The navigation system is controlled using a long range communication module bridged with a microcontroller to provide effective communication. The conveyer belt attached to the bot is conveyer belt where the motors are connected to the motor driver and on receiving signal from the controller it starts to rotate.
- [20] The segregation system built in this bot uses two sensors for its entire process. The infrared sensor is used to check the oncoming of the waste and initialize the segregation system.
- [21] The Inductive Proximity sensor is used to detect metallic waste. A servo motor is connected at the end to separate the waste based on the data from the inductive proximity sensor. The fig 1 describes the block diagram of the bot.

- [22] The water cleaning bot is an invention that is used to clean water surfaces. It eliminates the floating debris from the water surface.
- [23] The bot has a conveyer belt which continuously runs and collects the wastes. These wastes move further through the bot and passes to the segregation segment where it gets segregated into metal and non-metals.
- [24] After segregation process is completed, it is automatically collected into collection tray. The segregation system consists of the IR sensor, Inductive proximity sensor, servos for moving left and right direction.
- [25] The IR sensor is used to detect the presence of any waste. Once the presence of waste is detected, the segregation system is initialized and the Inductive Proximity sensor detects the incoming waste is a metal or non-metal.
- [26] Based on the result coming from the proximity sensor the servo moves left or right. The right compartment in the collection tray is for metallic wastes and left compartment is for non-metallic wastes.
- [27] The water cleaning bot is also used to monitor water quality and acquire information about the total dissolved solids, pH value, and turbidity of the water. These values will be collected using the corresponding sensors.
- [28] The navigation of the bot is controlled via Long Range (LoRa) communication protocol.
- [29] This brings to a head that water cleaning bot could be seen as an enhanced effective tool for preserving cleanness at water surfaces from debris.
- [30] The bot diverts this floating trash between metallic and non-metallic waste items. It possesses all those advanced sensors necessary and systems with high precision separation mechanism.
- [31] Thus, it supports efficiently in management of the rubbish. Other quality parameter measurements such as total dissolved solids, pH, or turbidity of water under which its utility for environmental conservation shows crucial.
- [32] This instrument is powerful and dependable in water surface management, as it makes use of the LoRa communication protocol for navigation. It substantially contributes to cleaner and healthier aquatic habitats.

BRIEF DESCRIPTION OF THE DRAWING(S)

The accompanying drawing, which is incorporated in and constitute a part of this disclosure, illustrate exemplary embodiment and, together with the description.

[33] **Figure 1** : illustrates the Block diagram of the bot

[34] **Figure 2** : illustrates the circuit diagram of the bot

[35] **Figure 3** : illustrates the interface of the android application

[36] **Figure 4** : illustrates the diagram of the model

ADVANTAGES OVER THE PRIOR ART

1. The energy utilized in working of the bot is a renewable source of energy which cuts the consumption of electricity and saves power.
2. Microcontroller code is easy and may be modified to the area of water body for completely autonomous navigation.
3. The bot requires just one human involvement for operation and control, reducing labor-intensive tasks.
4. This prototype offers a more cost-effective solution to reduce water pollution on a bigger scale.

CLAIMS

We Claim:

Claim 1 : *Efficiency in collecting wastes*

The water cleaning bot system is designed to efficiently collect floating debris, microplastics and pollutants from various water bodies comprising :

- a conveyer belt which continuously runs and collects the wastes;
- segregation segment where it gets segregated into metal and non-metals;
- collection tray to collect the segregated wastes, wherein right compartment is for metallic wastes and left compartment is for non-metallic wastes;
- Long Range (LoRa) communication protocol which substantially contributes to cleaner and healthier aquatic habitats.

Claim 2:

The segregation system as claimed in claim 1 comprising:

- the IR sensor which is used to detect the presence of any waste;
- inductive proximity sensor;
- servos for moving left and right direction.

Claim 3:

The water cleaning bot system is also used to monitor water quality and acquire information about the total dissolved solids, pH value, and turbidity of the water.

Claim 4: *Renewable energy integration*

The bot is powered using solar energy, thereby reducing operational costs and environmental impact, and enabling continuous and long-term operations.

Claim 5: *Modular design*

The bot is built with a modular structure, allowing for easy maintenance and the replacement of parts, sensors, and systems ensuring longevity and adaptability of the bot.

Claim 6 :*Real time Data collection*

The bot is equipped with water quality monitoring sensors, which provides real time data about the quality of the water that can be transmitted to monitoring sectors and offices for analysis and report.

Claim 7: *Minimal environmental impact*

The environmental impact is minimal because the material of the bot and operational mechanisms are built in such a way that it ensures minimal disruption to aquatic life while removing the wastes.

Claim 8 : *Cost-effective manufacturing*

The material used in making the bot is durable and strong yet it is cost-effective because of the design taken into consideration.

Dated this 29th day of June 2025.

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AGENT FOR THE APPLICANT(S)

“Design and Development of a Water Cleaning Bot for all water bodies”

ABSTRACT

The water cleaning bot is a novel solution to aquatic environment maintenance that combines modern waste management and water quality monitoring devices. The bot collects floating waste from water surfaces using a conveyor belt system, which is controlled via an Android interface. A sophisticated segregation system, which employs infrared (IR) and inductive proximity sensors, separates the collected trash into metallic and non-metallic components, which are then placed in specified compartments of a collecting tray. Furthermore, the bot is outfitted with sensors that monitor critical water quality metrics such as total dissolved solids (TDS), pH levels, and turbidity, giving useful data for environmental assessments. The Long Range (LoRa) protocol enables dependable navigation and communication over long distances. This technique provides a long-term and effective solution for maintaining water quality and reducing pollution in aquatic habitats.